A Preliminary Study on Identifying Hollow Cathode Operating Modes Using Image Recognition and Machine Learning

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Abstract: The hollow cathode, which provides ionization and plume neutralization, plays a critical role in electric propulsion system stability and lifetime. Its operating modes, spot and plume, directly affect performance, yet conventional classification methods rely on electrical signals, requiring intrusive probes and complex equipment. This study proposes a vision-based classification method using convolutional neural networks (CNNs) to address these limitations. Experiments were carried out with a heaterless argon hollow cathode operating in triode mode, under two mass flow rates (7 and 9 sccm) and five discharge currents (0.3–1.0 A). A labeled dataset of about 2100 images was collected, and four pretrained CNN models (AlexNet, VGG16-BN, ResNet101, and MobileNetV3) were tested. All models achieved 100% validation accuracy within 15 epochs. While deeper models converged faster and showed strong performance, lightweight models also maintained high accuracy. MobileNetV3, in particular, proved well suited for mobile deployment, balancing recognition accuracy with computational efficiency. This work provides the first demonstration of CNN-based image classification for hollow cathode mode recognition, offering a non-intrusive and intuitive diagnostic method. Future efforts will focus on improving generalization across different cathodes and extending the approach to other thruster plume modes, such as the jet and bell modes of Hall thrusters. The method has potential to reduce variability in human visual judgment and enable automated ground testing and health monitoring of electric propulsion systems.

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Nomenclature

Acc	=	Accuracy
CNN	=	Convolutional Neural Network
Epoch	=	One full pass of the dataset during training
GPU	=	Graphics Processing Unit
IAT	=	Ion acoustic turbulence
Loss	=	Loss function value
RMS	=	Root mean square
sccm	=	Standard cubic centimeters per minute (mass flow rate unit)
SGD	=	Stochastic gradient descent optimizer
V_a	=	Anode voltage (V)
$V_{a,p-p}$	=	Anode peak-to-peak voltage (V)
V_k	=	Keeper voltage (V)
$V_{k,p-p}$	=	Keeper peak-to-peak voltage (V)

I.Introduction

Electric propulsion has high efficiency and long lifetime, giving it clear advantages over chemical propulsion. It is now a key technology for present and future space missions. Among different types, electrostatic thrusters are widely used because they have high specific impulse and good thrust-to-power ratio. Ion thrusters and Hall thrusters are the most common [1]. The hollow cathode is a critical part for these types of thrusters, as it provides electrons to sustain the discharge and neutralize the ion beam. It directly affects ignition reliability, operational stability, and system lifetime[2].

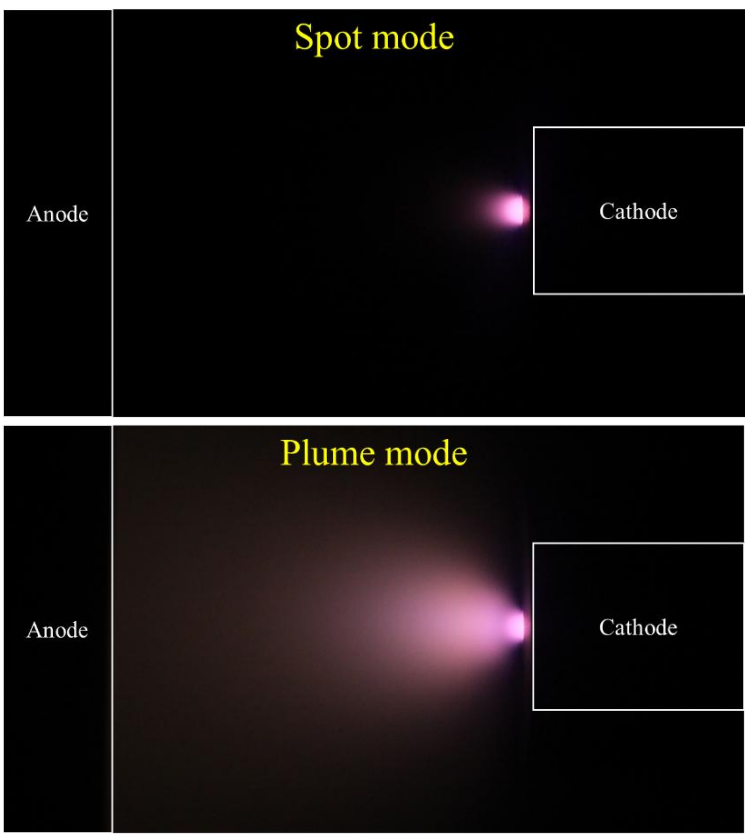

Figure 1. Hollow cathode spot and plume mode image

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The hollow cathode usually works in two modes (**Figure 1**). In spot mode, the discharge is stable, oscillations are small, and erosion is low. It shows a steady plasma ball downstream of the keeper orifice[3]. In plume mode, the discharge has stronger oscillations and more energetic ions[4]. The plasma forms a bright cone from the keeper to the anode, which causes faster wear. The change between the two modes is continuous and linked to ionization instabilities and ion acoustic turbulence (IAT)[5]. It is often marked by higher discharge voltage and stronger oscillations[6]. Most classification methods use electrical signals such as peak-to-peak voltage or current oscillation[7, 8]. Although these methods are practical, they rely on empirical laws and require intrusive probes, which may disturb the system, necessitate complex equipment, and potentially lead to human misjudgment.

To address these limitations, non-intrusive imaging provides a promising alternative. Unlike electrical diagnostics, imaging can directly capture the spatial structure of the discharge in a simple and disturbance-free way[9]. With recent advances in computer vision, convolutional neural networks (CNNs) have become powerful tools for image recognition because they can automatically extract and learn specific features from raw images[10]. By applying CNN-based classification to hollow cathode images, this study develops a mode recognition framework that complements traditional diagnostics and supports more reliable and efficient monitoring of plasma behavior[11, 12].

This study proposes a hollow cathode mode recognition framework using convolutional neural networks (CNNs). This approach fills the gap of traditional diagnostics, simplifies ground testing, and links visible plume patterns with plasma processes. In the future, it has potential to support real-time cathode health monitoring and thruster plume classification.

II. Hollow cathode experimental setup

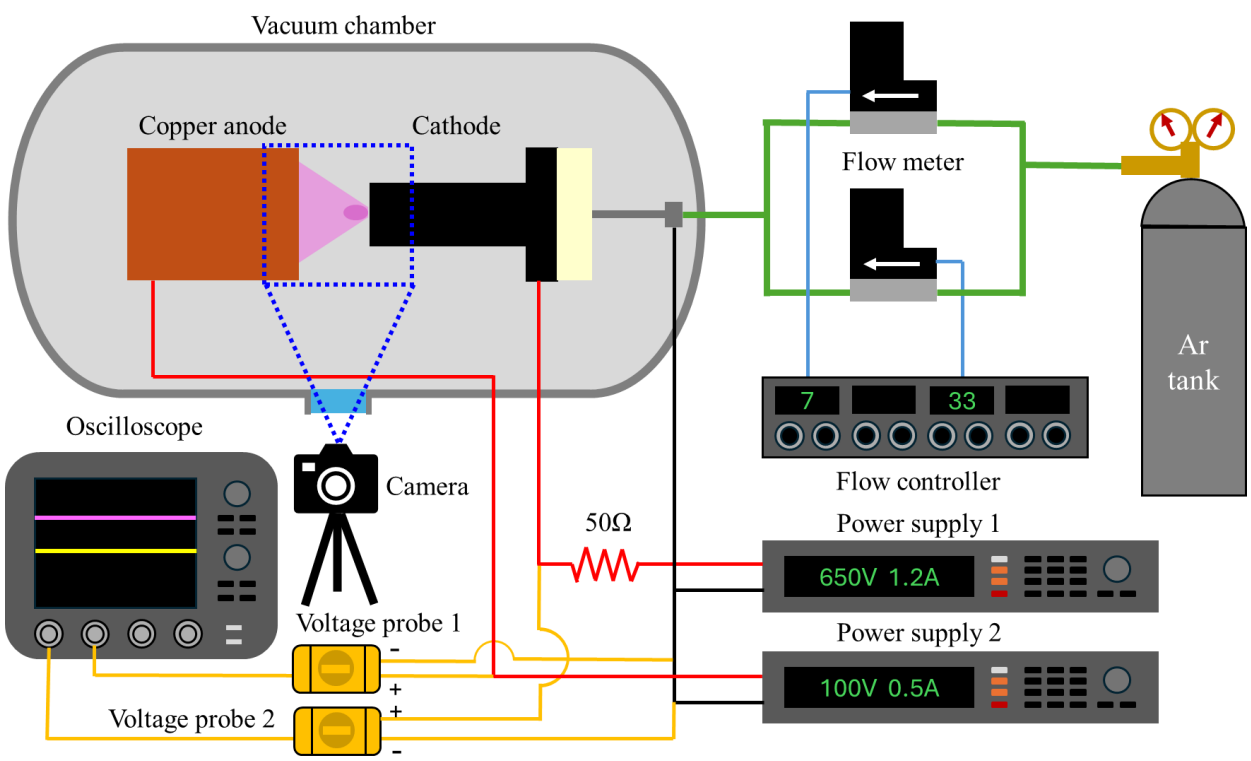

Figure 2. Experimental setup for cathode testing

The heaterless argon hollow cathode used in this study[13] was developed in our laboratory and tested under ten different operating conditions. The test matrix included two mass flow rates (7 and 9 sccm) and five discharge currents

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(0.3, 0.4, 0.6, 0.8, and 1.0 A). The experimental setup for cathode testing is illustrated in **Figure 2**. During ignition, the cathode was operated at 650 V / 1.2 A, with the mass flow gradually reduced from 40 sccm to the target value. After ignition, the keeper power supply remained on, the cathode operated in triode mode[14]. For each operating condition, 200 discharge images were recorded using a fixed visible-light camera (Fujifilm X-H2 with FUJINON XF80mmF2.8 R LM OIS WR Macro lens). In total, a dataset of 2100 labeled images was collected, with 50% at 1x magnification, 25% at 1.4x, and 25% at 2x.

A cylindrical copper anode (diameter 80 mm, length 100 mm) was placed 30 mm downstream of the keeper exit plane. Two power supplies (BK Precision MR100020) were used to drive the keeper and anode separately. Argon flow was controlled by two mass flow controllers (Brooks Instruments 5850E), with maximum capacities of 10 sccm and 50 sccm. During ignition, the 50 sccm controller provided high flow, while the 10 sccm controller supplied steady flow for stable operation. The propellant was high-purity argon (5N2 grade). Voltage measurements were performed with a four-channel, 12-bit oscilloscope (TELEDYNE LECROY WaveSurfer 4024HD), recording the voltage–time waveforms of the anode and keeper. Cybertek DP6700A and Pintek DP-15K probes were used, with a measurement range up to 7 kV DC, bandwidths of 100 MHz and 75 MHz, and accuracy within ±2%. All experiments were carried out in a vacuum chamber (1.5 m × 1 m) at a background pressure of 1.0–1.2 × 10^{-4} torr.

III.Dataset Construction and Preprocessing

Several empirical formulas have been proposed to distinguish spot mode from plume mode. Sary et al. suggested using the amplitude of anode current oscillations, where plume mode is identified when the standard deviation of the anode current exceeds 9% of its mean[15, 16]. The predator–prey model provides another criterion based on the ratio of discharge current to mass flow rate[17]. When this ratio exceeds characteristic constantly, the cathode operates in plume mode. This model is especially useful for predicting plume mode under high current and low flow conditions. Georgin et al. considered instabilities related to ion acoustic turbulence (IAT) and proposed an empirical formula that combines IAT frequency with cathode geometry parameters, applicable at high discharge currents[18]. Sengupta et al. recommended using anode voltage as an indicator, where plume mode is empirically observed when the voltage rises by more than 5 V[19].

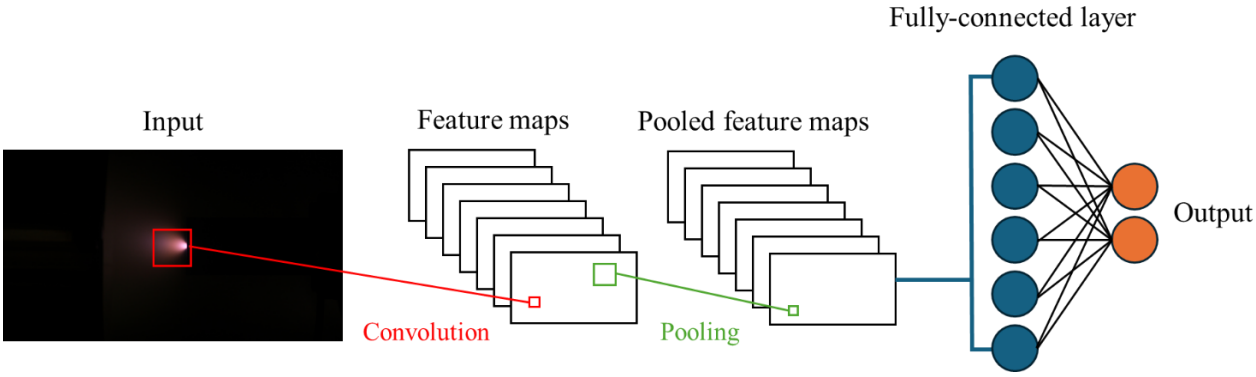

Figure 3. Schematic of hollow cathode image classification process using a convolutional neural network

In this study, mode classification is based on the peak-to-peak voltage of the anode ($V_{a,p-p}$) and keeper ($V_{k,p-p}$). When the voltage difference is greater than 5 V, the discharge is classified as plume mode; when lower, it is classified as spot mode, the mean anode voltage (V_a) and keeper voltage (V_k) has also been monitored. For image classification, several pretrained CNN models were used, including AlexNet, VGG16-BN, ResNet101, and MobileNetV3. A CNN is a deep learning architecture inspired by the human visual cortex. It consists of convolution, pooling, and fully connected layers, and it extracts local features through convolution layers, reduces data dimensions while keeping essential information with pooling layers, and performs classification through fully connected layers, as shown in **Figure 3**. When the dataset is small, transfer learning is often applied by using models pretrained on large datasets such as ImageNet. This reduces training time and data requirements and improves performance under limited data[20]. The pretrained models describe as below, AlexNet was a milestone in deep learning, showing the strength of CNNs

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in the 2012 ImageNet competition, with five convolutional and three fully connected layers[10]. VGG16 improved representation ability by stacking multiple 3×3 filters, while VGG16-BN added batch normalization to speed up convergence and improve stability[21, 22]. The ResNet series introduced residual blocks with shortcut connections, which solved the vanishing gradient problem in deep networks and allowed very deep models to be trained effectively[23]. ResNet101 is deep model with 101 layers and a bottleneck design, making it well suited for large-scale classification and high-level feature learning. MobileNetV3, developed by Google, is optimized for mobile and resource-limited environments[24].

The hardware environment included an NVIDIA GeForce RTX 5070 Ti, running on Windows 10, with Python as the development platform. Since CNN models contain a large number of parameters, training was accelerated with Graphics Processing Unit (GPU) computing[25]. Before inputting the images into the CNN model, preprocessing was applied. All images were resized to 224 × 224 pixels to meet the model input requirement. The dataset contained about 2100 images, randomly divided into 70% for training and 30% for validation. Data augmentation was applied before training: each image had a 50% chance of horizontal flipping and was randomly shifted by 10% along both the x and y axes. The training used a batch size of 4, a learning rate of 0.001, and stochastic gradient descent (SGD) with a momentum of 0.9. The loss function was cross-entropy loss, combined with a learning rate scheduler that reduced the learning rate by a factor of 0.9 every three epochs to improve convergence..

IV.Result & Discussion

A. Hollow cathode operating situation

The discharge voltage and current relationship was first measured at fixed flow rates, as shown in **Figure 4**, with the pink dashed line indicating the mode classification threshold. The operating flow rates were 7 sccm and 9 sccm, and the two voltage–current curves were similar, likely because the flow difference was small, and the plasma behavior remained comparable. In both cases, the peak-to-peak values of the anode and keeper increased with discharge current, following similar trends. At around 0.5 A, both exceeded 5 V and showed a sharp increase, indicating a mode transition. At higher discharge currents, the peak-to-peak values at 7 sccm were larger than those at 9 sccm, suggesting that lower flow makes the discharge more sensitive to current[26]. The average anode voltage was affected by the mode transition and rose significantly at the transition point. In contrast, the average keeper voltage was not influenced by the transition and showed a steady decrease with increasing discharge current.

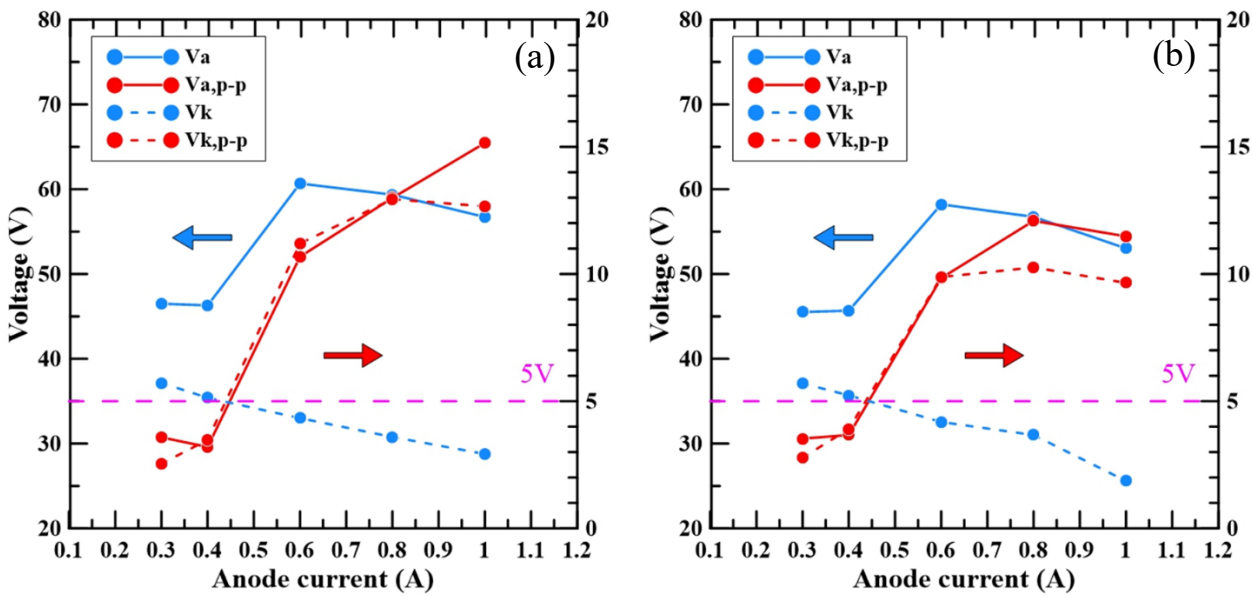

Figure 4. Voltage–current characteristics of the hollow cathode at different flow rates: (a) 7 sccm, (b) 9 sccm

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B. Performance Comparison of CNN networks

The following **Figure 5-10** show the training and validation Loss/Accuracy curves of different models. In most cases, the training loss and validation loss decreased together, suggesting limited signs of overfitting in this task. For deeper CNN models, such as VGG16-BN and ResNet101, the validation accuracy approached unity (≈100%) in the first epoch, suggesting that this task is relatively simple for deep networks. In contrast, AlexNet and MobileNetV3 showed oscillations in the early stage. For AlexNet, we attribute this behavior to its early design without batch normalization or residual connections, which makes optimization more sensitive to learning-rate and momentum settings, especially under small mini-batches[22], while MobileNetV3 uses depthwise separable convolutions enables a lightweight architecture, but may increase sensitivity to hyperparameter tuning[24], which we observed to cause initial instability in our experiments. However, the validation accuracy of both models eventually stabilized, indicating that early oscillations did not affect the final performance.

The results show that all four models used in this study (AlexNet, VGG16-BN, ResNet101, and MobileNetV3) achieved very high accuracy in classifying hollow cathode operating modes, reaching 100% validation accuracy within 15 epochs. The fast convergence and high accuracy are likely due to the binary nature of the task and the uniform dark background, which removed distractions and made feature extraction straightforward. Since all models reached perfect classification on both the training and validation sets with no errors, the confusion matrices were identical. Therefore, only one representative confusion matrix (**Figure 9**) is presented for reference.

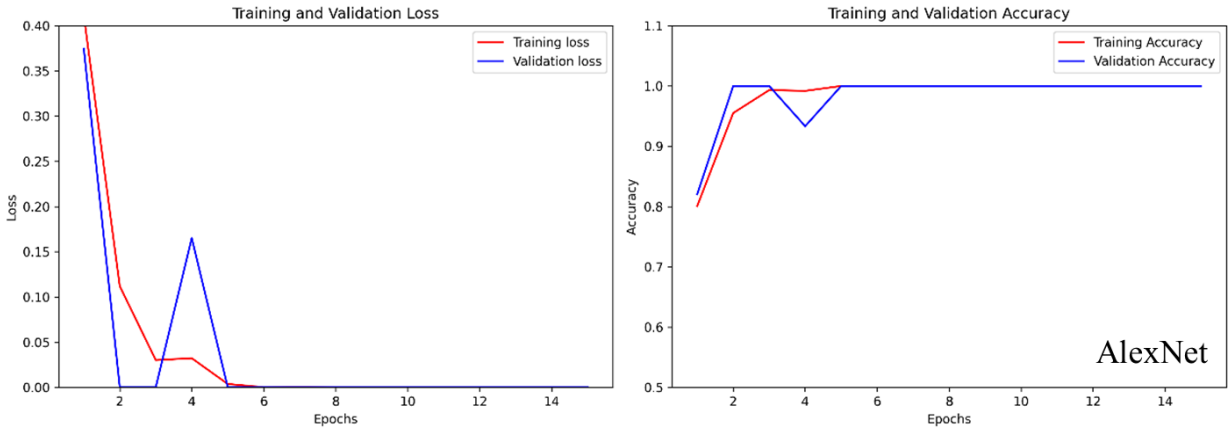

Figure 5. Training and validation loss and accuracy curves of the AlexNet model

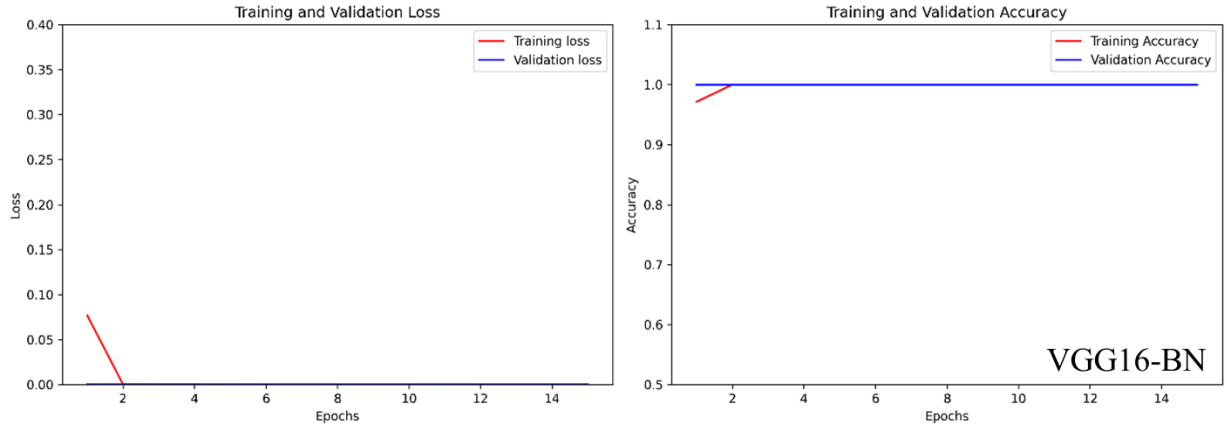

Figure 6. Training and validation loss and accuracy curves of the VGG16-BN model

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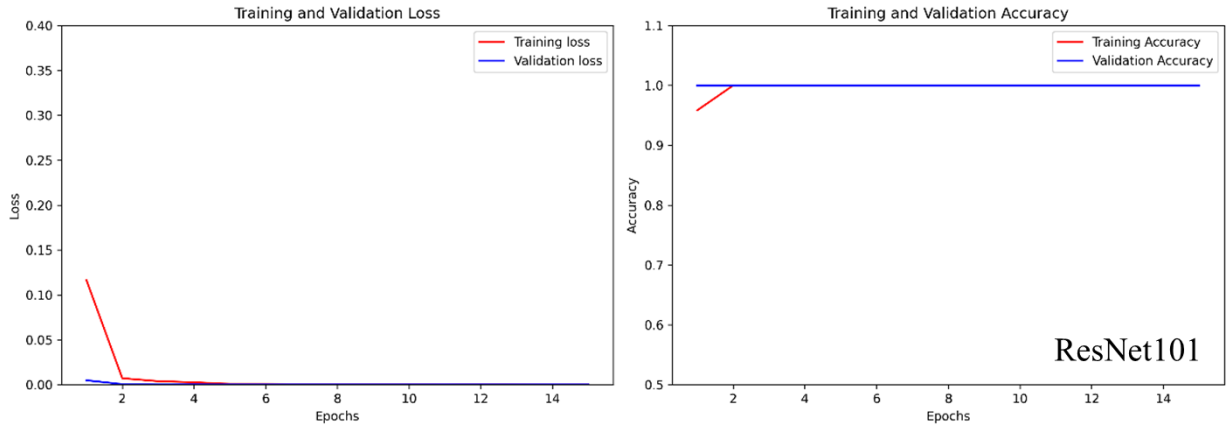

Figure 7. Training and validation loss and accuracy curves of the Resnet101 model

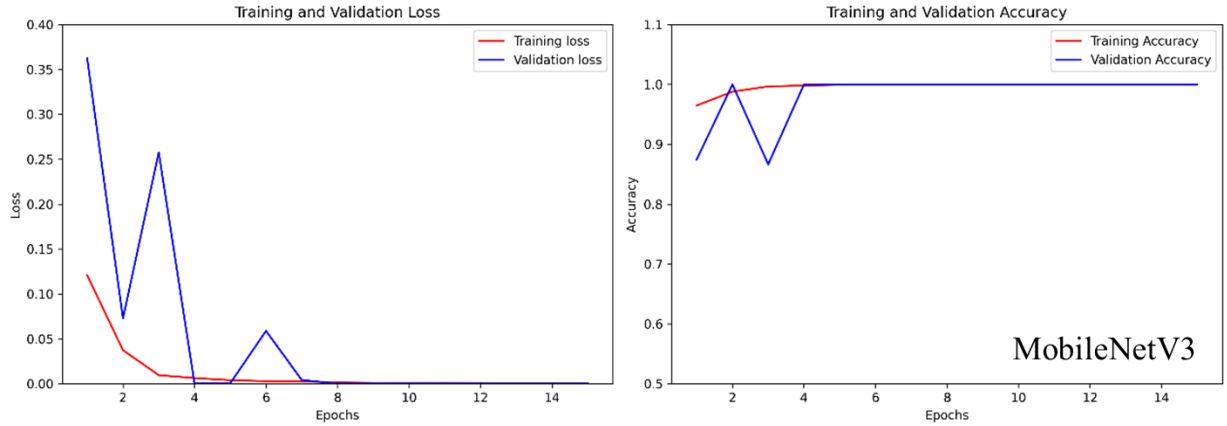

Figure 8. Training and validation loss and accuracy curves of the MobileNetV3 model

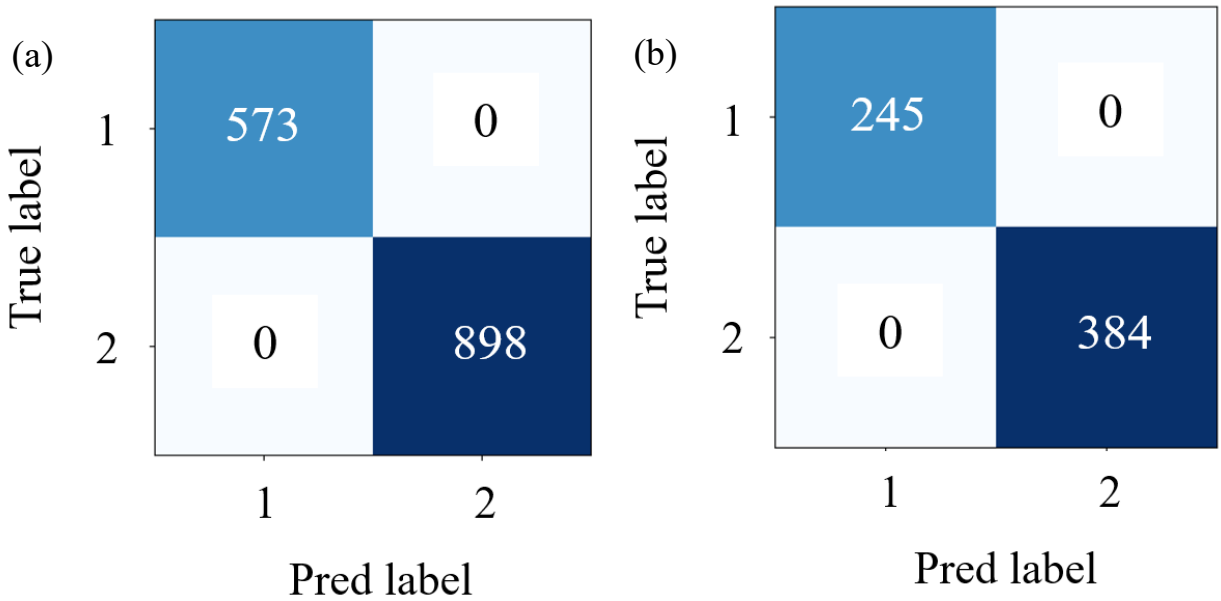

Figure 9. Confusion matrices of hollow cathode mode classification: (a) training set, (b) validation set

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The average training time per epoch for each model is shown in **Table 1**. VGG16-BN required significantly longer time due to its deeper layers, larger number of parameters, and the added computation from batch normalization. ResNet101 has a depth of 101 layers with a large number of residual blocks. This greatly increases both computation and memory usage, making its training time comparable to that of VGG16-BN.

To apply this classification method in ground tests of electric propulsion systems, usability on mobile devices is an important consideration. Among the tested models, MobileNetV3 was designed by Google specifically for mobile and embedded systems[24]. It is lightweight and can be easily deployed on frameworks such as TensorFlow Lite or PyTorch Mobile. Experimental results show that small models are sufficient for this type of task, MobileNetV3 achieves good classification accuracy while greatly reducing model size and computational cost, making it highly suitable for practical deployment[27, 28].

Table 1. Average training time per epoch for different convolutional neural network models

CNN model	Average time (s)
AlexNet	3.63
VGG16-BN	12.83
ResNet101	11.57
MobileNetV3	6.33

V.Future work

One focus of future work is to improve the generalization of the model to different cathodes. Images of various cathodes will be collected under the same procedure for different modes and then tested with the trained model to ensure it is not limited to a single cathode. Another focus is to verify whether this approach can be extended to other thruster discharge modes, such as the jet mode and bell mode of Hall thrusters, which show distinct plume characteristics. If this method can be applied and extended to plume classification, it will reduce inconsistencies caused by human visual judgment and improve the reliability of mode identification.

VI.Conclusion

This study proposed a vision-based approach using convolutional neural networks (CNNs) to classify hollow cathode operating modes (spot and plume) and demonstrated its feasibility. The argon hollow cathode operated in triode mode, with mode transition occurring at a discharge current of about 0.5 A for flow rates of 7 sccm and 9 sccm. Currents above 0.5 A corresponded to plume mode, while those below 0.5 A corresponded to spot mode. The classification criterion was whether the peak-to-peak voltage of the keeper and anode exceeded 5 V.

Four CNN models (AlexNet, VGG16-BN, ResNet101, and MobileNetV3) were tested, and after parameter adjustments, all reached 100% validation accuracy within 15 epochs. This result is partly due to the binary nature of the task and the uniform dark background. The task was relatively simple for deeper models such as VGG16-BN and ResNet101, suggesting that lighter models can be used to save computational resources without compromising accuracy. For practical ground testing and deployment on mobile devices, MobileNetV3 is recommended, as it was specifically optimized for such platforms and provides reliable classification while greatly reducing model size and computational load.

Acknowledgments

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